

Generate Realistic Fake Data in Colab Using Faker & Mimesis

Create massive synthetic datasets for testing, demos and machine learning!

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1 Install Libraries & Sample Repo

```
!pip install Faker==2.0.5 -q
!npx degit RealImpactAnalytics/trumania/trumania trumania
```

We install Faker for generating fake data and clone the trumania sample repo.

2 Generate Data with Faker (100,000 rows)

```
from faker import Faker
import pandas as pd
import random
fake = Faker()

def create_rows_faker(num=1):
    output = [{"name":fake.name(),
               "address":fake.address(),
               "email":fake.email(),
               "city":fake.city(),
               "state":fake.state(),
               "date_time":fake.date_time(),
               "randomdata":random.randint(1000,2000)}
                  for x in range(num)]
    return output

df_faker = pd.DataFrame(create_rows_faker(100000))
```

Sample of Faker Data

	name	email	city	state	date_time	randomdata
1	John Doe	johndoe@example.com	Chicago	IL	2021-07-21 10:23:45	1532
2	Mary Smith	marysmith@example.com	Austin	TX	2021-06-11 22:15:10	1899
3	David Brown	davidbrown@example.com	Boston	MA	2021-02-18 09:30:01	1201
4	Jessica Lee	jessicalee@example.com	New York	NY	2021-03-05 14:05:33	1774
5	Michael Johnson	michael@example.com	Seattle	WA	2021-01-30 17:45:55	1660

Great for quick & simple synthetic data generation!

3 Generate Data with Mimesis (1,000,000 rows)

```
from mimesis import Person, Address, Datetime, locales
from mimesis.enums import Gender
import pandas as pd, random

person = Person('en')
address = Address(locales.EN_AU)
datetime = Datetime()

def create_rows_mimesis(num=1):
    output = [{"name":person.full_name(gender=Gender.FEMALE),
               "address":address.address(),
               "email":person.email(),
               "blood_type":person.blood_type(),
               "sity":address.city(),
               "state":address.state(),
               "last_login":datetime.datetime(),
               "height":person.height(),
               "weight":person.weight(),
               "random_data":random.randint(1000,100000+1000)}
                  for x in range(num)]
    return output

df_mimesis = pd.DataFrame(create_rows_mimesis(1000000))
df_mimesis.to_csv('mimesis_data.csv', index=False)
```

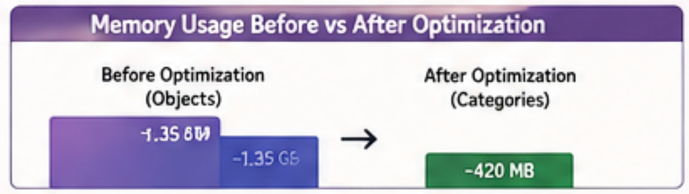
Sample of Mimesis Data

	name	email	city	state	blood_type	height	weight	last_login	random_data
1	Olivia Harris	olivia.harris@example.com	Sydney	NSW	O+	172	65	2023-08-01 14:12:33	85234
2	Emily Johnson	emily.johnson@example.com	Melbourne	VIC	A-	168	58	2023-08-02 09:21:10	64312
3	Charlotte Smith	charlotte.smith@example.com	Brisbane	QLD	B+	163	54	2023-08-02 17:05:42	91345
4	Amelia Brown	amelia.brown@example.com	Perth	WA	AB+	170	62	2023-08-03 11:33:21	22514
5	Sophia Wilson	sophia.wilson@example.com	Adelaide	SA	O-	165	57	2023-08-03 19:44:08	78120

Mimesis is fast and generates highly realistic & diverse data!

4 Clean & Optimize Data

```
df_mimesis.state = df_mimesis.state.astype('category')
df_mimesis.city = df_mimesis.city.astype('category')
df_mimesis.blood_type = df_mimesis.blood_type.astype('category')
df_mimesis.height = pd.to_numeric(df_mimesis.height,
                                  errors='coerce')
```



Using categories & numeric types reduces memory and improves performance!

5 Dataset Overview

```
df_mimesis.info()
```

Output (after optimization)

```
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 11 columns):
 #   Column       Non-Null Count    Dtype
---  ------       --------------    -----
 0   name         1000000 non-null  object
 1   address      1000000 non-null  object
 2   email        1000000 non-null  object
 3   blood_type   1000000 non-null  category
 4   city         1000000 non-null  category
 5   state        1000000 non-null  category
 6   last_login   1000000 non-null  datetime64[ns]
 7   height       1000000 non-null  float64
 8   weight       1000000 non-null  float64
 9   random_data  1000000 non-null  int64
dtypes: category(3), datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 420.2+ MB
```

1 Million rows of realistic synthetic data ready to use!

Why use Faker & Mimesis?

- **Safe** — No real data exposure
- **Fast** — Generate large datasets quickly
- **Realistic** — High quality synthetic data
- **Flexible** — Perfect for tests, demos & ML

Export & Use

Saved to CSV and ready for:

- Data Analysis
- Machine Learning
- API Testing
- Application Development

Tips

- Faker is simple & lightweight – great for basic fake data.
- Mimesis is powerful & configurable – ideal for large, complex datasets.
- Always optimize data types for memory efficiency!
- Use to_csv() to save and share your synthetic datasets.

